

ALK fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes ALK gene fusions from the msk_impact_50k_2026 study, covering 272 fusion events. 226/272 fusions (83.1%) were in-frame. 261/272 fusions (96.0%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test ($p=0.00191661$, statistically significant at $\alpha=0.05$). No literature benchmark is configured for ALK.

1 Results summary

1.1 Fusion partner frequency

Fusion-partner frequency was tabulated across 272 analyzed fusion events, identifying 32 distinct fusion partner genes. The most frequent partner was EML4, observed in 221 events.

1.2 Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 222/226 (98.2%) of in-frame fusions retained the domain, $p=0.00191661$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.000999001.

1.3 Domain disruption

Domain-disruption analysis was skipped: ALK has no disruption_required_domains configured; domain-disruption analysis was skipped.

1.4 Cutpoint detection

Cutpoint detection scanned 271 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 972 aa (permutation-corrected $p=0.000999001$, statistically significant at $\alpha=0.05$). This is 12 aa from the nearest configured domain boundary at 960 aa.

1.5 Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=261$) and "not_retained" ($n=10$). For Frame_status, retained: 222/227 (97.8%, 95% CI 94.9%-99.1%); not_retained: 4/5 (80.0%, 95% CI 37.6%-96.4%). Welch's t-test comparing

Tumor_variant_count between groups gave means of 65.6139 vs 31.8, $p=0.00432792$ (statistically significant at $\alpha=0.05$).

1.6 Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

1.7 Exon retention

Exon retention reported: No target exon was supplied in params or GeneConfig.

1.8 Joint partner

Joint partner reported: ALK has no gene_pair configured; joint-partner analysis was skipped.

2 Tables

2.1 composite_score composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
EML4	221	0.8125	0.30004340774793187		0.8865878486549185	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.6503541781991878	1
LRP1B	1	0.003676470588235294	0.30004340774793187	1.0		0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3683101330469789	2
EMILIN1	1	0.003676470588235294	0.30004340774793187		0.9559412550066756	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.36004911836073056	3
CLTC	2	0.007352941176470588	0.30004340774793187		0.931909212283044	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3569217868206379	4
FN1	1	0.003676470588235294	0.30004340774793187		0.931909212283044	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.35554311035004965	5
NOTCH1	1	0.003676470588235294	0.30004340774793187		0.931909212283044	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.35554311035004965	6
NRP2	1	0.003676470588235294	0.30004340774793187		0.931909212283044	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.35554311035004965	7
STRN	7	0.025735294117647058	0.30004340774793187		0.8865153538050734	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.35530382070895955	8
KIF5B	5	0.01838235294117647	0.30004340774793187		0.8945260347129506	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3540484704380101	9
PLEKHA7	1	0.003676470588235294	0.30004340774793187		0.9158878504672897	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3525391050095957	10
SPTBN1	4	0.014705882352941176	0.30004340774793187		0.8895193591455274	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.35173104229853	11
RANBP2	3	0.011029411764705883	0.30004340774793187		0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.34953878104823544	12
PICALM	3	0.011029411764705883	0.30004340774793187		0.8811748998664887	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.348787779713122	13
SQSTM1	2	0.007352941176470588	0.30004340774793187		0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.34816010457764723	14
TPM1	2	0.007352941176470588	0.30004340774793187		0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.34816010457764723	15
CTNNA1	1	0.003676470588235294	0.30004340774793187		0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.346781428107059	16
DCTN1	1	0.003676470588235294	0.30004340774793187		0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.346781428107059	17
DIAPH2	1	0.003676470588235294	0.30004340774793187		0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.346781428107059	18

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty	Components_applicable	Composite_score	Rank
DYSF	1	0.003676470588235294	0.30004340774793187		0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.346781428107059	19
GCC2	1	0.003676470588235294	0.30004340774793187		0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.346781428107059	20
HMBOX1	1	0.003676470588235294	0.30004340774793187		0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.346781428107059	21
PLEKHH2	1	0.003676470588235294	0.30004340774793187		0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.346781428107059	22
PPP1CB	1	0.003676470588235294	0.30004340774793187		0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.346781428107059	23
TFG	1	0.003676470588235294	0.30004340774793187		0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.346781428107059	24
TRAF3	1	0.003676470588235294	0.30004340774793187		0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.346781428107059	25
TTC27	1	0.003676470588235294	0.30004340774793187		0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.346781428107059	26
VCL	1	0.003676470588235294	0.30004340774793187		0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.346781428107059	27
ZFPM2	1	0.003676470588235294	0.30004340774793187		0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.346781428107059	28
H6PD	1	0.003676470588235294	0.30004340774793187		0.7503337783711616	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3214977164915717	29
CTSE	1	0.003676470588235294	0.30004340774793187		0.39118825100133514	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.2541579301097292	30
SOS1	1	0.003676470588235294	0.30004340774793187			0.6853431332412958	['confidence_certainty', 'domain_retention', 'recurrence']	0.2225355483655125	31
ASAP2	1	0.003676470588235294	0.30004340774793187		0.0	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.1808101330469789	32

2.2 cutpoint_detection cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
223	0	261	1	9	0.0	0.03690036900369004	1.4329692908744058
516	0	261	2	8	0.0	0.001230012300123001	2.9100905455940684
785	0	261	3	7	0.0	3.658029145347215e-05	4.436752838604533
939	2	259	3	7	0.0003515999093716071	0.0003515999093716071	3.4539512455419974
972	2	259	4	6	1.3820564716971664e-05	1.3820564716971664e-05	4.8594742110534295
1023	10	251	4	6	0.0007961806348620647	0.0007961806348620647	3.098988389769786
1028	11	250	4	6	0.05976095617529881	0.0010657190672512412	2.9723572639982643
1032	12	249	4	6	0.066	0.0013947454739763106	2.8555050392241372
1035	13	248	4	6	0.07228915662650602	0.0017901726849544938	2.747105073765603
1040	14	247	4	6	0.07862903225806452	0.002258991897970247	2.646085326703862
1042	16	245	4	6	0.08502024291497975	0.003444927806265043	2.4628198749665864
1045	18	243	4	6	0.09795918367346938	0.005008607520231548	2.300282998750876
1047	19	242	4	6	0.11111111111111111	0.005949364209574388	2.225529443519691

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
1051	20	241	4	6	0.12448132780082988	0.007005060123461933	2.154588132872754
1052	22	239	4	6	0.13807531380753138	0.00948736346530418	2.0228544610771384
1058	254	7	10	0	0.0	1.0	-0.0
1059	255	6	10	0	0.0	1.0	-0.0
1061	258	3	10	0	0.0	1.0	-0.0
1064	259	2	10	0	0.0	1.0	-0.0
1070	260	1	10	0	0.0	1.0	-0.0

2.3 domain_retention frame_domain_contingency_table

222	39
4	6

2.4 domain_retention domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained
kinase	Protein kinase domain	271	261	0	10	0.0

2.5 frequency Partner_gene_counts

Partner_gene	Event_count
ASAP2	1
CLTC	2
CTNNA1	1
CTSE	1
DCTN1	1
DIAPH2	1
DYSF	1
EMILIN1	1
EML4	221
FN1	1
GCC2	1
H6PD	1
HMBOX1	1
KIF5B	5
LRP1B	1
NOTCH1	1
NRP2	1
PICALM	3
PLEKHA7	1
PLEKHH2	1
PPP1CB	1
RANBP2	3
SOS1	1
SPTBN1	4
SQSTM1	2
STRN	7
TFG	1
TPM1	2

Partner_gene	Event_count
TRAF3	1
TTC27	1
VCL	1
ZFPM2	1

2.6 Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_pos	protein_neg	domain_pos	domain_neg	main_retained	domain_pos	domain_neg	disrupted_domains	description	annotation	effect_on_fusion
EVT-P-0024679-T01-IM6-39	P-0024679-T01-IM6		ALK-CTSE	CTSE	unknown	five_prime	29543594	7	516	True	lost	0.0	False		Protein kinase domain			ALK (NM_004304) - CTSE(NM_001910) Rearrangement: t(1;2)(q32.1;p23.3)(chr1:g.206318648; chr2:g.29543594)	NA	
EVT-P-0020387-T01-IM6-40	P-0020387-T01-IM6		ALK-DCTN1	DCTN1	in-frame	three_prime	29446896	20	1058	True	retained	1.0	False		Protein kinase domain			Antisense Fusion DCTN1 (NM_004082) - ALK (NM_004304) rearrangement: c.3196+238DCTN1_c.3173-502:ALKdel Protein Fusion: in frame {DCTN1:ALK} ALK (NM_004304) - DYSF (NM_001130987) rearrangement: c.3172+909ALK_c.3402+7348:DYSFdup	NA	
EVT-P-0006776-T01-IM5-44	P-0006776-T01-IM5		ALK-DYSF	DYSF	unknown	five_prime	29447418	19	1058	True	lost	0.0	False		Protein kinase domain			Antisense fusion EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1490-639:EML4_c.3173-252:ALKinv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0031827-T02-IM6-45	P-0031827-T02-IM6		ALK-EML4	EML4	in-frame	three_prime	29446646	20	1058	True	retained	1.0	False		Protein kinase domain					

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	breakpoint_4	main_ref	donor_frame	acceptor_frame	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0010041-T01-IM5-55	P-0010041-T01-IM5		ALK- EML4	EML4	in-frame	three_prime	29447942	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fuses with ALK exons 20-29): c.701-6977:EML4_c.3172+385:ALKInv	NA	
EVT-P-0008332-T01-IM5-56	P-0008332-T01-IM5		ALK- EML4	EML4	unknown	five_prime	29446912	20	1058	True	lost	0.0	False		Protein kinase domain		EML4 (NM_019063) - ALK (NM_004304) rearrangement: c.1489+828:EML4_c.3173-518:ALKdup	NA	
EVT-P-0010177-T01-IM5-57	P-0010177-T01-IM5		ALK- EML4	EML4	in-frame	three_prime	29447358	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused in frame with ALK exons 20-29): c.1490-2649:EML4_c.3173-964:ALKInv	NA	
EVT-P-0007705-T01-IM5-58	P-0007705-T01-IM5		ALK- EML4	EML4	unknown	three_prime	29447417	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 with ALK exons 20-29): c.1594:EML4_c.3172+910:ALKInv	NA	
EVT-P-0010978-T01-IM5-59	P-0010978-T01-IM5		ALK- EML4	EML4	out-of-frame	three_prime	29446414	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fused with ALK exons 20-29): c.2154+2237:EML4_c.3173-20:ALKInv	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	breakpoint_4	is_main_ref	is_disrupted	description	annotation	effect_on_fra
EVT-P-0012679-T01-IM5-65	P-0012679-T01-IM5		ALK- EML4	EML4	in-frame	three_prime	29447466	19	1058	True	retained	1.0	False	Protein kinase domain	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 with ALK exons 20-29): c.1489+2812:EML4_c.3172+861:ALKInv	
EVT-P-0012778-T01-IM5-66	P-0012778-T01-IM5		ALK- EML4	EML4	in-frame	three_prime	29447442	19	1058	True	retained	1.0	False	Protein kinase domain	EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+1320:EML4_c.3172+885:ALKInv	
EVT-P-0012778-T02-IM6-67	P-0012778-T02-IM6		ALK- EML4	EML4	in-frame	three_prime	29447442	19	1058	True	retained	1.0	False	Protein kinase domain	EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.493+1320:EML4_c.3172+885:ALKInv	
EVT-P-0007211-T01-IM5-68	P-0007211-T01-IM5		ALK- EML4	EML4	in-frame	three_prime	29447489	19	1058	True	retained	1.0	False	Protein kinase domain	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29) : c.667+165:EML4_c.3172+838:ALKInv	
EVT-P-0007125-T01-IM5-69	P-0007125-T01-IM5		ALK- EML4	EML4	unknown	three_prime	29446597	20	1058	True	retained	1.0	False	Protein kinase domain	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-14 with ALK exons 20-29) : c.1328:EML4_c.3173-203:ALKInv	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	breakpoint_4	is_main_ref	is_disrupted	is_disrupted	is_disrupted	is_disrupted	is_disrupted	description	annotation	effect_on_fusion
EVT-P-0006950-T02-IM6-70	P-0006950-T02-IM6		ALK-EML4	EML4	in-frame	three_prime	29447384	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 with ALK exons 20-29): c.1489+2302: EML4_c.3172+943: ALKinv	NA	
EVT-P-0013202-T01-IM5-71	P-0013202-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29447178	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29): c.667+2565: EML4_c.3173-784: ALKinv	NA	
EVT-P-0006950-T01-IM5-72	P-0006950-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29447384	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 with ALK exons 20-29): c.1489+2302: EML4_c.3172+943: ALKinv	NA	
EVT-P-0013453-T01-IM5-73	P-0013453-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29446597	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fused in-frame with ALK exons 20-29): c.2056+551: EML4_c.3173-203: ALKinv	NA	
EVT-P-0013864-T01-IM5-74	P-0013864-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29448209	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exons 1 to 13 with exons 20-29 of ALK): c.1489+658: EML4_c.3172+118: ALKinv	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	protein_retained	breakpoint_4	main_retained	domain_disrupted	domain_disrupted	domain_disrupted	domain_disrupted	description_annotation	effect_on_fusion
EVT-P-0013869-T01-IM5-75	P-0013869-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29446712	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exons 1 to 13 with exons 20-29 of ALK) : c.1490-187:EML4_c.3173-318:ALKInv Protein fusion: in frame (EML4-ALK) EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused in-frame to ALK exons 20-29) : c.667+2294-285:ALKInv Protein fusion: in frame (EML4-ALK) EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exons 20-29) : c.667+6128-2147:EML4_c.3172+203:ALKInv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0014270-T01-IM6-76	P-0014270-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446679	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused in-frame to ALK exons 20-29) : c.667+2294-285:ALKInv Protein fusion: in frame (EML4-ALK) EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exons 20-29) : c.667+6128-2147:EML4_c.3172+203:ALKInv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0014310-T01-IM6-77	P-0014310-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447544	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exons 20-29) : c.667+6128-2147:EML4_c.3172+203:ALKInv Protein fusion: in frame (EML4-ALK) EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 with ALK exons 20-29) : c.1490-2147:EML4_c.3172+203:ALKInv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0006370-T01-IM5-78	P-0006370-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29448124	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 with ALK exons 20-29) : c.1490-2147:EML4_c.3172+203:ALKInv Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	is_main_ref	is_cds	is_intron	is_exon	is_disrupted	description	annotation	effect_on_fra
EVT-P-0015434-T02-IM6-86	P-0015434-T02-IM6		ALK- EML4	EML4	in-frame	three_prime	294448197	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon 13 fused to ALK exon20): c.1489+63:EML4_c.3172+130:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 NA (NM_001145076) - ALK (NM_004304) Fusion (EML4 exon 13 fused with ALK exon 20) : c.2068+156:EML4_c.3173- 693:ALKInv Protein fusion: in frame (EML4- ALK) EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.668- 119:EML4_c.3172+520:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 NA (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK ex- ons 20-29): c.667+516:EML4_c.3173- 415:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 NA (NM_019063) - ALK (NM_004304)) fusion (EML4 exon 6fused to ALK exon 20): c.668- 6666:EML4_c.3173- 132:ALKInv Protein Fusion: in frame {EML4:ALK}		
EVT-P-0005321-T01-IM5-87	P-0005321-T01-IM5		ALK- EML4	EML4	in-frame	three_prime	294448250	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon 13 fused with ALK exon 20) : c.2068+156:EML4_c.3173- 693:ALKInv Protein fusion: in frame (EML4- ALK) EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.668- 119:EML4_c.3172+520:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 NA (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK ex- ons 20-29): c.667+516:EML4_c.3173- 415:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 NA (NM_019063) - ALK (NM_004304)) fusion (EML4 exon 6fused to ALK exon 20): c.668- 6666:EML4_c.3173- 132:ALKInv Protein Fusion: in frame {EML4:ALK}		
EVT-P-0029971-T01-IM6-88	P-0029971-T01-IM6		ALK- EML4	EML4	in-frame	three_prime	294447807	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.668- 119:EML4_c.3172+520:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 NA (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK ex- ons 20-29): c.667+516:EML4_c.3173- 415:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 NA (NM_019063) - ALK (NM_004304)) fusion (EML4 exon 6fused to ALK exon 20): c.668- 6666:EML4_c.3173- 132:ALKInv Protein Fusion: in frame {EML4:ALK}		
EVT-P-0015921-T03-IM6-89	P-0015921-T03-IM6		ALK- EML4	EML4	in-frame	three_prime	29446809	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK ex- ons 20-29): c.667+516:EML4_c.3173- 415:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 NA (NM_019063) - ALK (NM_004304)) fusion (EML4 exon 6fused to ALK exon 20): c.668- 6666:EML4_c.3173- 132:ALKInv Protein Fusion: in frame {EML4:ALK}		
EVT-P-0015961-T01-IM6-90	P-0015961-T01-IM6		ALK- EML4	EML4	in-frame	three_prime	29446526	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon 6fused to ALK exon 20): c.668- 6666:EML4_c.3173- 132:ALKInv Protein Fusion: in frame {EML4:ALK}		

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	breakpoint_4	main_ref	donor_frame	inframe	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0015961-T02-IM6-91	P-0015961-T02-IM6		ALK-EML4	EML4	in-frame	three_prime	29446526	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.668-6666:EML4_132:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0015961-T03-IM6-92	P-0015961-T03-IM6		ALK-EML4	EML4	in-frame	three_prime	29446526	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.668-6666:EML4_132:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0031827-T01-IM6-93	P-0031827-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446700	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK fusion: c.1490-673:EML4_306:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0027119-T01-IM6-94	P-0027119-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446711	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK fusion (EML4 exons 1-12 fused to ALK exons 20-29): c.1490-837:EML4_317:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0029750-T01-IM6-95	P-0029750-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446660	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1490-2574:EML4_266:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	is_main_ref	is_disrupted	is_inframe	is_retained	is_deleted	description	annotation	effect_on_fra
EVT-P-0029550-T01-IM6-96	P-0029550-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	294447863	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.494-721:EML4_c.3172+464:ALK	NA	Inv
EVT-P-0016280-T01-IM6-97	P-0016280-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	294447905	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) rearrangement: c.1316-2732:EML4_c.3172+422:ALK	NA	Inv
EVT-P-0016580-T01-IM6-98	P-0016580-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447579	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1489+2257:EML4_c.3172+748:ALK	NA	Inv
EVT-P-0005305-T01-IM5-99	P-0005305-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29447764	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29): c.668-5757:EML4_c.3172+563:ALK	NA	Inv
EVT-P-0016960-T01-IM6-100	P-0016960-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447838	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.667+7453:EML4_c.3172+489:ALK	NA	Inv

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	breakpoint_4	is_main_ref	is_disrupted	is_disrupted	is_disrupted	is_disrupted	is_disrupted	description	annotation	effect_on_fusion
EVT-P-0017592-T01-IM5-101	P-0017592-T01-IM5		ALK- EML4	EML4	in-frame	three_prime	29447548	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exon 1-5 fused with ALK exon 20-29): c.494-5963:EML4 Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.1315+641:EML4_c.3172+779:ALKInv	NA	
EVT-P-0029400-T01-IM6-102	P-0029400-T01-IM6		ALK- EML4	EML4	in-frame	three_prime	29447668	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1315+641:EML4_c.3172+659:ALKInv	NA	
EVT-P-0004218-T01-IM5-103	P-0004218-T01-IM5		ALK- EML4	EML4	in-frame	three_prime	29447087	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exon 20 fused with ALK exon 20) : c.2068+156:EML4_c.3173-693:ALKInv	NA	
EVT-P-0017978-T01-IM6-104	P-0017978-T01-IM6		ALK- EML4	EML4	in-frame	three_prime	29446609	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.667+7306:EML4_c.3173-215:ALKInv	NA	
EVT-P-0017978-T02-IM6-105	P-0017978-T02-IM6		ALK- EML4	EML4	in-frame	three_prime	29446609	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.667+7306:EML4_c.3173-215:ALKInv	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	protein_retained	breakpoint_4	main_ref	donor_frame	main_ref	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0019491-T01-IM6-113	P-0019491-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446596	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 with ALK exons 20-29): c.1490-486:EML4_c.3173-202:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0019513-T01-IM6-114	P-0019513-T01-IM6		ALK-EML4	EML4	unknown	three_prime	29448403	19	1032	False	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 19-29): c.1489+2278:EML4_c.3096:ALKInv Protein Fusion: mid-exon {EML4:ALK}	NA	
EVT-P-0019533-T01-IM6-115	P-0019533-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446494	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29): c.667+6837:EML4_c.3173-100:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0009102-T01-IM5-116	P-0009102-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29446513	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused in frame with ALK exons 20-29, which include the ALK kinase domain): c.1490-244:EML4_c.3173-119:ALKInv Protein fusion: in frame (EML4-ALK)	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	protein_retained	breakpoint_4	main_ref	donor_frame	main_ref	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0028901-T01-IM6-117	P-0028901-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446607	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1316-844:EML4_c.3173-213:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0019786-T01-IM6-118	P-0019786-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447575	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fused in-frame with ALK exons 20-29): c.2056+82:EML4_c.3172+752:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0019972-T01-IM6-119	P-0019972-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29448101	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exon 20-29): c.493+1179:EML4_c.3172+226:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0020124-T01-IM6-120	P-0020124-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447955	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused in-frame with ALK exons 20-29): c.1489+879:EML4_c.3172+372:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0020138-T01-IM6-121	P-0020138-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447907	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.668-3016:EML4_c.3172+420:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	breakpoint_4	main_ref	donor_frame	maintained	domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0003383-T01-IM5-122	P-0003383-T01-IM5		ALK- EML4	EML4	in-frame	three_prime	29447572	19	1058	True	retained	1.0	False	Protein kinase domain			NA Protein fusion: in frame (EML4-ALK)	NA	
EVT-P-0020408-T01-IM6-123	P-0020408-T01-IM6		ALK- EML4	EML4	in-frame	three_prime	29446779	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1489+1227:385:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	EML4_c.3173-
EVT-P-0003323-T01-IM5-124	P-0003323-T01-IM5		ALK- EML4	EML4	in-frame	three_prime	29447530	19	1058	True	retained	1.0	False	Protein kinase domain			NA Protein fusion: in frame (EML4-ALK)	NA	
EVT-P-0003297-T01-IM5-125	P-0003297-T01-IM5		ALK- EML4	EML4	in-frame	three_prime	29448259	19	1058	True	retained	1.0	False	Protein kinase domain			NA Protein fusion: in frame (EML4-ALK)	NA	
EVT-P-0020584-T01-IM6-126	P-0020584-T01-IM6		ALK- EML4	EML4	in-frame	three_prime	29447048	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused to ALK exon 20-29): c.209-4846:EML4_654:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	EML4_c.3173-
EVT-P-0020780-T01-IM6-127	P-0020780-T01-IM6		ALK- EML4	EML4	in-frame	three_prime	29447084	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1 - 13 fused to ALK exons 20 - 29): c.1489+1278:690:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	EML4_c.3173-
EVT-P-0020825-T03-IM6-128	P-0020825-T03-IM6		ALK- EML4	EML4	in-frame	three_prime	29446767	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-2 fused to ALK exons 20-29):c.208+4373:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	EML4_c.3173-

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	breakpoint_4	main_ref	clonally_expressed	in_frame	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0022101-T01-IM6-135	P-0022101-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447759	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exon18 fused with ALK exon 20): c.2057-510:EML4_c.2057-510:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused with ALK exons 20-29): c.2333:EML4_c.3173-190:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+107:EML4_c.3172+470:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) -ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1490-2055:EML4_c.3173-536:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1523-2563:EML4_c.3172:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0022138-T01-IM6-136	P-0022138-T01-IM6		ALK-EML4	EML4	unknown	three_prime	29446584	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused with ALK exons 20-29): c.2333:EML4_c.3173-190:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+107:EML4_c.3172+470:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) -ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1490-2055:EML4_c.3173-536:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1523-2563:EML4_c.3172:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0036247-T01-IM6-137	P-0036247-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447857	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+107:EML4_c.3172+470:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1490-2055:EML4_c.3173-536:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1523-2563:EML4_c.3172:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0036106-T01-IM6-138	P-0036106-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446930	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) -ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1490-2055:EML4_c.3173-536:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1523-2563:EML4_c.3172:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0028745-T01-IM6-139	P-0028745-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447509	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1523-2563:EML4_c.3172:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	protein_lost	breakpoint_4	main_ref	donor_frame	inframe	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0022398-T01-IM6-140	P-0022398-T01-IM6		ALK- EML4	EML4	in-frame	three_prime	29448035	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1- 5 fused with ALK exons 20- 29): c.494- 7853:EML4_	NA c.3172+292:ALKInv	
EVT-P-0002266-T01-IM3-141	P-0002266-T01-IM3		ALK- EML4	EML4	in-frame	five_prime	29447288	20	1058	True	lost	0.0	False		Protein kinase domain			Fusion: in frame {EML4:ALK} ALK (NM_004304) and EML4 (NM_019063) reciprocal inversion (13026566bp): c.3173- 889:ALK_c.	NA 208+1022:EML4inv	
EVT-P-0002266-T01-IM3-142	P-0002266-T01-IM3		ALK- EML4	EML4	in-frame	three_prime	29447283	20	1058	True	retained	1.0	False	Protein kinase domain				NA Protein fusion: in frame (ALK- EML4) NA	NA	
EVT-P-0028541-T03-IM6-143	P-0028541-T03-IM6		ALK- EML4	EML4	in-frame	three_prime	29446922	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_001145076) - ALK (NM_004304) fusion: c.208+4631	NA EML4_c.3173-	
EVT-P-0022926-T01-IM6-144	P-0022926-T01-IM6		ALK- EML4	EML4	in-frame	three_prime	29446558	20	1058	True	retained	1.0	False	Protein kinase domain				fusion: c.493+983:EML4_c.3173- 164:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK ex- ons 20-29): c.493+983:EML4_c.3173- 164:ALKInv Protein Fusion: in frame {EML4:ALK}	NA NA	
EVT-P-0035121-T01-IM6-145	P-0035121-T01-IM6		ALK- EML4	EML4	in-frame	three_prime	29447383	19	1058	True	retained	1.0	False	Protein kinase domain				fusion: c.668- 4857:EML4_	NA c.3172+944:ALKInv	
EVT-P-0002217-T02-IM3-146	P-0002217-T02-IM3		ALK- EML4	EML4	in-frame	three_prime	29446594	20	1058	True	retained	1.0	False	Protein kinase domain				fusion: c.668- 4857:EML4_	NA NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	protein_retained	breakpoint_4	main_ref	donor_frame	main_ref	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0023368-T01-IM6-147	P-0023368-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446713	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29 including the kinase domain) : c.667+7004:319:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) inversion: c.2242+31:739:ALKInv Protein fusion: in frame (EML4-ALK) EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 19-29): c.667+4775:389:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.668-4498:EML4 Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) Inversion: c.1315+2666: EML4_c.3172+122:ALKInv	NA	
EVT-P-0002024-T01-IM3-148	P-0002024-T01-IM3		ALK-EML4	EML4	in-frame	three_prime	29447133	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) inversion: c.2242+31:739:ALKInv Protein fusion: in frame (EML4-ALK) EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 19-29): c.667+4775:389:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.668-4498:EML4 Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) Inversion: c.1315+2666: EML4_c.3172+122:ALKInv	NA	EML4_c.3173-
EVT-P-0023962-T01-IM6-149	P-0023962-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29448820	19	1023	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 19-29): c.667+4775:389:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.668-4498:EML4 Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) Inversion: c.1315+2666: EML4_c.3172+122:ALKInv	NA	EML4_c.3173-
EVT-P-0024136-T01-IM6-150	P-0024136-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29448266	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.668-4498:EML4 Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) Inversion: c.1315+2666: EML4_c.3172+122:ALKInv	NA	EML4_c.3068-
EVT-P-0001811-T01-IM3-151	P-0001811-T01-IM3		ALK-EML4	EML4	in-frame	three_prime	29448205	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) Inversion: c.1315+2666: EML4_c.3172+122:ALKInv	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_pos	breakpoint_pos	breakpoint_pos	protein_pos	breakpoint_pos	main_ref	donor_pos	acceptor_pos	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0028541-T02-IM6-152	P-0028541-T02-IM6		ALK-EML4	EML4	in-frame	three_prime	29446922	20	1058	True	retained	1.0	False	Protein kinase domain		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused with ALK exons 20-29): c.208+4631-528:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-2 fused to ALK exons 20-29): c.209-990:EML4_c.3173-483:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused to ALK exons 20-29): c.2243-217:EML4_c.3173-120:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon13 fused with ALK exons 20) : c.1641+555:EML4_c.3213:ALKInv	NA (5076)	
EVT-P-0024907-T01-IM6-153	P-0024907-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446877	20	1058	True	retained	1.0	False	Protein kinase domain		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused to ALK exons 20-29): c.209-990:EML4_c.3173-483:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused to ALK exons 20-29): c.2243-217:EML4_c.3173-120:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon13 fused with ALK exons 20) : c.1641+555:EML4_c.3213:ALKInv	NA	
EVT-P-0034569-T02-IM6-154	P-0034569-T02-IM6		ALK-EML4	EML4	in-frame	three_prime	29446514	20	1058	True	retained	1.0	False	Protein kinase domain		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-20 fused to ALK exons 20-29): c.2243-217:EML4_c.3173-120:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon13 fused with ALK exons 20) : c.1641+555:EML4_c.3213:ALKInv	NA	
EVT-P-0034450-T01-IM6-155	P-0034450-T01-IM6		ALK-EML4	EML4	unknown	three_prime	29446354	20	1071	False	retained	1.0	False	Protein kinase domain		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exon13 fused with ALK exons 20) : c.1641+555:EML4_c.3213:ALKInv	NA	
EVT-P-0001532-T01-IM3-156	P-0001532-T01-IM3		ALK-EML4	EML4	in-frame	three_prime	29447553	19	1058	True	retained	1.0	False	Protein kinase domain		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exon13 fused with ALK exons 20) : c.1641+555:EML4_c.3213:ALKInv	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint	breakpoint	breakpoint	protein_is_in_frame	protein_is_in_frame	main_retained	donor_insertion	protein_is_in_frame	protein_is_in_frame	disrupted_domains	domains_annotation	effect_on_frame	
EVT-P-0028541-T01-IM6-157	P-0028541-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446922	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused with ALK exons 20-29): c.208+4631-528:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-18 fused with ALK exons 20-29): c.1981-2450:EML4_c.3173:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fuses with ALK exons 20-29) : c.667+7374:EML4_c.3172+876:ALKInv Protein Fusion: in frame {ALK:EML4} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fuses with ALK exons 20-29) : c.2312:EML4_c.3173-558:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exons 20-29): c.2312:EML4_c.3173-558:ALKInv Protein Fusion: mid-exon {EML4:ALK}	NA (5076)	2
EVT-P-0026407-T01-IM6-158	P-0026407-T01-IM6		ALK-EML4	EML4	unknown	three_prime	29446392	20	1059	False	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-18 fused with ALK exons 20-29): c.1981-2450:EML4_c.3175:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fuses with ALK exons 20-29) : c.667+7374:EML4_c.3172+876:ALKInv Protein Fusion: in frame {ALK:EML4} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fuses with ALK exons 20-29) : c.2312:EML4_c.3173-558:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exons 20-29): c.2312:EML4_c.3173-558:ALKInv Protein Fusion: mid-exon {EML4:ALK}	NA (5076)	2
EVT-P-0026650-T02-IM6-159	P-0026650-T02-IM6		ALK-EML4	EML4	in-frame	five_prime	29447451	19	1058	True	lost	0.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fuses with ALK exons 20-29) : c.667+7374:EML4_c.3172+876:ALKInv Protein Fusion: in frame {ALK:EML4} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fuses with ALK exons 20-29) : c.2312:EML4_c.3173-558:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exons 20-29): c.2312:EML4_c.3173-558:ALKInv Protein Fusion: mid-exon {EML4:ALK}	NA (5076)	2
EVT-P-0026657-T01-IM6-160	P-0026657-T01-IM6		ALK-EML4	EML4	unknown	three_prime	29446952	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-20 fuses with ALK exons 20-29) : c.2312:EML4_c.3173-558:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exons 20-29): c.2312:EML4_c.3173-558:ALKInv Protein Fusion: mid-exon {EML4:ALK}	NA (5076)	2
EVT-P-0026657-T03-IM6-161	P-0026657-T03-IM6		ALK-EML4	EML4	unknown	three_prime	29446952	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exons 20-29): c.2312:EML4_c.3173-558:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exons 20-29): c.2312:EML4_c.3173-558:ALKInv Protein Fusion: mid-exon {EML4:ALK}	NA (5076)	2

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_coding	in_frame	main_retention	fusion_retention	protein_coding	in_frame	main_retention	fusion_retention	disrupted_domains	annotation	effect_on_fusion
EVT-P-0015152-T02-IM6-162	P-0015152-T02-IM6		ALK-EML4	EML4	in-frame	three_prime	29447079	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused in-frame to ALK exons 20-29): c.2242+127:685:ALKInv Protein fusion: in frame (EML4:ALK)	NA	
EVT-P-0032862-T01-IM6-163	P-0032862-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447488	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exon 11 fused to ALK exon20): c.1489+671:839:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0027119-T03-IM6-164	P-0027119-T03-IM6		ALK-EML4	EML4	in-frame	three_prime	29446711	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1316-837:EML4_c.3173-317:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0032683-T01-IM6-165	P-0032683-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447763	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2122:EML4_c.3172+564:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0028339-T01-IM6-166	P-0028339-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447680	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused with ALK exons 20-29): c.208+1245:647:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	is_main_ref	is_in_frame	is_retained	is_lost	is_disrupted	description	annotation	effect_on_fra
EVT-P-0032518-T01-IM6-167	P-0032518-T01-IM6		ALK- EML4	EML4	unknown	three_prime	29448359	19	1047	False	retained	1.0	False	Protein kinase domain				EML4 (NM_001145076) - ALK (NM_004304) fusion: c.1315+2607:EML4_c.3140:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused to ALK ex- ons 20-29): c.2243- 157:EML4_c.3172+593:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exons 1-5 with ALK exons 20-29) : c.668- 1726:EML4_c.3173- 124:ALKInv Protein fusion: in frame (EML4- ALK) NA NA Protein fusion: in frame (EML4- ALK) GCC2 (NM_181453) - ALK (NM_004304) rearrange- ment : c.4448- 1230:GCC2_c.3173- 132:ALKInv Protein fusion: in frame (GCC2- ALK) ALK (NM_004304) rearrange- ment: t(1;2)(p36.22;p23.2)(chr1:g.9314150::chr	NA	2
EVT-P-0027853-T01-IM6-168	P-0027853-T01-IM6		ALK- EML4	EML4	in-frame	three_prime	29447734	19	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused to ALK ex- ons 20-29): c.2243- 157:EML4_c.3172+593:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exons 1-5 with ALK exons 20-29) : c.668- 1726:EML4_c.3173- 124:ALKInv Protein fusion: in frame (EML4- ALK) NA NA Protein fusion: in frame (EML4- ALK) GCC2 (NM_181453) - ALK (NM_004304) rearrange- ment : c.4448- 1230:GCC2_c.3173- 132:ALKInv Protein fusion: in frame (GCC2- ALK) ALK (NM_004304) rearrange- ment: t(1;2)(p36.22;p23.2)(chr1:g.9314150::chr	NA	2
EVT-P-0015110-T01-IM6-169	P-0015110-T01-IM6		ALK- EML4	EML4	in-frame	three_prime	29446518	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exons 1-5 with ALK exons 20-29) : c.668- 1726:EML4_c.3173- 124:ALKInv Protein fusion: in frame (EML4- ALK) NA NA Protein fusion: in frame (EML4- ALK) GCC2 (NM_181453) - ALK (NM_004304) rearrange- ment : c.4448- 1230:GCC2_c.3173- 132:ALKInv Protein fusion: in frame (GCC2- ALK) ALK (NM_004304) rearrange- ment: t(1;2)(p36.22;p23.2)(chr1:g.9314150::chr	NA	2
EVT-P-0003446-T01-IM5-170	P-0003446-T01-IM5		ALK- EML4	EML4	in-frame	three_prime	29446534	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exons 1-5 with ALK exons 20-29) : c.668- 1726:EML4_c.3173- 124:ALKInv Protein fusion: in frame (EML4- ALK) NA NA Protein fusion: in frame (EML4- ALK) GCC2 (NM_181453) - ALK (NM_004304) rearrange- ment : c.4448- 1230:GCC2_c.3173- 132:ALKInv Protein fusion: in frame (GCC2- ALK) ALK (NM_004304) rearrange- ment: t(1;2)(p36.22;p23.2)(chr1:g.9314150::chr	NA	2
EVT-P-0004516-T02-IM5-171	P-0004516-T02-IM5		ALK- GCC2	GCC2	in-frame	three_prime	29446526	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exons 1-5 with ALK exons 20-29) : c.668- 1726:EML4_c.3173- 124:ALKInv Protein fusion: in frame (EML4- ALK) NA NA Protein fusion: in frame (EML4- ALK) GCC2 (NM_181453) - ALK (NM_004304) rearrange- ment : c.4448- 1230:GCC2_c.3173- 132:ALKInv Protein fusion: in frame (GCC2- ALK) ALK (NM_004304) rearrange- ment: t(1;2)(p36.22;p23.2)(chr1:g.9314150::chr	NA	2
EVT-P-0035712-T01-IM6-172	P-0035712-T01-IM6		ALK- H6PD	H6PD	unknown	five_prime	29462508	13	785	True	lost	0.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exons 1-5 with ALK exons 20-29) : c.668- 1726:EML4_c.3173- 124:ALKInv Protein fusion: in frame (EML4- ALK) NA NA Protein fusion: in frame (EML4- ALK) GCC2 (NM_181453) - ALK (NM_004304) rearrange- ment : c.4448- 1230:GCC2_c.3173- 132:ALKInv Protein fusion: in frame (GCC2- ALK) ALK (NM_004304) rearrange- ment: t(1;2)(p36.22;p23.2)(chr1:g.9314150::chr	NA	2

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	is_main_ref	is_disrupted	is_disrupted	disrupted_domain	disrupted_domain	description_annotation	annotation_site	effect_on_fusion
EVT-P-0018541-T01-IM6-174	P-0018541-T01-IM6		ALK-LRP1B	LRP1B	in-frame	five_prime	29450289	17	972	True	lost	0.0	False			Protein kinase domain		ALK (NM_004304) - LRP1B (NM_018557) Rearrangement: c.2914+151:ALK_c.344-85984:LRP1Bdup Protein Fusion: in frame {ALK:LRP1B}	NA	
EVT-P-0027092-T01-IM6-176	P-0027092-T01-IM6		ALK-PLEKHH2	PLEKHH2	in-frame	three_prime	29446506	20	1058	True	retained	1.0	False			Protein kinase domain		PLEKHH2 (NM_172069) - ALK (NM_004304) rearrangement: c.502+545:PLEKHH2_c.3173-112:ALKinv Protein Fusion: in frame {ALK:LRP1B}	NA	
EVT-P-0027363-T02-IM6-177	P-0027363-T02-IM6		ALK-RANBP2	RANBP2	in-frame	three_prime	29447364	19	1058	True	retained	1.0	False			Protein kinase domain		RANBP2 (NM_006267) - ALK (NM_004304) rearrangement: c.2603-1556:RANBP2_c.3172+963:ALKinv Protein Fusion: in frame {PLEKHH2:ALK}	NA	
EVT-P-0068865-T02-IM7-178	P-0068865-T02-IM7		ALK-RANBP2	RANBP2	in-frame	five_prime	29447373	19	1058	True	lost	0.0	False			Protein kinase domain		ALK (NM_004304) - RANBP2 (NM_006267) rearrangement: c.3172+954:ALK_c.2602+1532:RANBP2 Protein Fusion: in frame {RANBP2:ALK}	NA	
EVT-P-0008843-T01-IM5-179	P-0008843-T01-IM5		ALK-RANBP2	RANBP2	in-frame	three_prime	29446488	20	1058	True	retained	1.0	False			Protein kinase domain		ALK (NM_004304) - RANBP2 (NM_006267) rearrangement: c.2603-1070:RANBP2_c.3173-94:ALKinv Protein fusion: in frame {ALK:RANBP2}	NA	
EVT-P-0015699-T01-IM6-181	P-0015699-T01-IM6		ALK-SPTBN1	SPTBN1	in-frame	three_prime	29447437	19	1058	True	retained	1.0	False			Protein kinase domain		SPTBN1 (NM_003128) - ALK (NM_004304) fusion (SPTBN1 exon 7 fused to ALK exon 20): c.764-1028:SPTBN1_c.3172+890:ALKinv Protein fusion: in frame (SPTBN1-ALK)	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1_pos	protein_2_pos	breakpoint_4	is_main_ref	is_in_frame	is_retained	is_in_domain	disrupted_domains	description_annotation	effect_on_fra
EVT-P-0016193-T05-IM6-182	P-0016193-T05-IM6		ALK-STRN	STRN	unknown	three_prime	29448326	19	1058	True	retained	1.0	False	Protein kinase domain				STRN (NM_003162) - ALK (NM_004304) fusion (STRN exons 1-3 fused with ALK exons 20-29): c.412+4919:STRN_c.3172+1:ALKdel	NA
EVT-P-0016193-T02-IM6-183	P-0016193-T02-IM6		ALK-STRN	STRN	unknown	three_prime	29448326	19	1058	True	retained	1.0	False	Protein kinase domain				ALK (NM_004304) - STRN (NM_003162):c.3172+1:ALK_c.412+4919:STRNdel	NA
EVT-P-0016193-T01-IM6-184	P-0016193-T01-IM6		ALK-STRN	STRN	unknown	three_prime	29448326	19	1058	True	retained	1.0	False	Protein kinase domain				ALK (NM_004304) - STRN (NM_003162):c.3172+1:ALK_c.412+4919:STRNdel	NA
EVT-P-0041438-T01-IM6-185	P-0041438-T01-IM6		ALK-TTC27	TTC27	out-of-frame	five_prime	29447817	19	1058	True	lost	0.0	False	Protein kinase domain				rearrangement: c.3172+510:ALK_c.538-4165:TTC27inv	NA
EVT-P-0006617-T01-IM5-187	P-0006617-T01-IM5		ASAP2-ALK	ASAP2	unknown	five_prime	30142753	1	223	True	lost	0.0	False	Protein kinase domain				Fusion: out of frame {ALK:TTC27} ASAP2 (NM_003887) - ALK (NM_004304) rearrangement event : c.127-20148:ASAP2_c.667+106:ALKdel	NA
EVT-P-0062994-T02-IM7-188	P-0062994-T02-IM7		CLTC-ALK	CLTC	in-frame	three_prime	29449624	18	1023	True	retained	1.0	False	Protein kinase domain				antisense fusion CLTC (NM_004859) - ALK (NM_004304) fusion: t(2;17)(p23;q23.1)(chr2:g.29449624::chr2:g.2944962	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_ratio	breakpoint_4	main_ref	donor_frame	main_ref	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0040574-T01-IM6-223	P-0040574-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29447738	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon5 fused with ALK exon20) : c.668-1589:EML4_c.3172+589:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0040576-T01-IM6-224	P-0040576-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29447321	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon5 fused with ALK exon20) : c.668-821:EML4_c.3173-927:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0040600-T01-IM6-225	P-0040600-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29447688	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+1795:EML4_c.3172+639:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0032683-T02-IM7-226	P-0032683-T02-IM7		EML4-ALK	EML4	in-frame	three_prime	29447763	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2122:EML4_c.3172+564:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0014385-T06-IM7-227	P-0014385-T06-IM7		EML4-ALK	EML4	in-frame	three_prime	29447676	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+1890:EML4_c.3172+651:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0067069-T01-IM7-228	P-0067069-T01-IM7		EML4-ALK	EML4	in-frame	three_prime	29448169	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+2305:EML4_c.3172+158:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0014385-T02-IM6-229	P-0014385-T02-IM6		EML4-ALK	EML4	in-frame	three_prime	29447676	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+1890:EML4_c.3172+651:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	is_in_frame	is_retained	is_disrupted	description	annotation	effect_on_fra
EVT-P-0012913-T02-IM6-237	P-0012913-T02-IM6		EML4-ALK	EML4	in-frame	three_prime	29446965	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.209-4133:EML4_c.3173-571:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-4017:EML4_c.3172+814:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.2254:EML4_c.3172+226:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.339-2119:EML4_c.3118:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-4308:EML4_c.3173-550:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+3829:EML4_c.3172+563:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+7821:EML4_c.3172+310:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0043350-T01-IM6-238	P-0043350-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29447513	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-4017:EML4_c.3172+814:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.2254:EML4_c.3172+226:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.339-2119:EML4_c.3118:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-4308:EML4_c.3173-550:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+3829:EML4_c.3172+563:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+7821:EML4_c.3172+310:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0043363-T01-IM6-239	P-0043363-T01-IM6		EML4-ALK	EML4	unknown	three_prime	29448101	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.2254:EML4_c.3172+226:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.339-2119:EML4_c.3118:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-4308:EML4_c.3173-550:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+3829:EML4_c.3172+563:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+7821:EML4_c.3172+310:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0043594-T01-IM6-240	P-0043594-T01-IM6		EML4-ALK	EML4	unknown	three_prime	29448381	19	1040	False	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.339-2119:EML4_c.3118:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-4308:EML4_c.3173-550:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+3829:EML4_c.3172+563:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+7821:EML4_c.3172+310:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0069396-T02-IM7-241	P-0069396-T02-IM7		EML4-ALK	EML4	in-frame	three_prime	29446944	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-4308:EML4_c.3173-550:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+3829:EML4_c.3172+563:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+7821:EML4_c.3172+310:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0069535-T01-IM7-242	P-0069535-T01-IM7		EML4-ALK	EML4	in-frame	three_prime	29447764	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+3829:EML4_c.3172+563:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+7821:EML4_c.3172+310:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0069617-T01-IM7-243	P-0069617-T01-IM7		EML4-ALK	EML4	in-frame	three_prime	29448017	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+7821:EML4_c.3172+310:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_pos	breakpoint_4	domain_retained	domain_lost	domain_deleted	disrupted_domains	sequence_annotation	clinical_effect_on_fra
EVT-P-0044432-T01-IM6-244	P-0044432-T01-IM6		EML4-ALK	EML4	unknown	three_prime	29448344	19	1052	False	retained	1.0	False	Protein kinase domain		EML4 (NM_019063) - ALK (NM_004304) fusion: c.2057-576:EML4_c.3155:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 NA (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 with ALK exons 20-29) : c.2243-181:EML4_c.3172+658:ALKInv Protein fusion: in frame (EML4-ALK) EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.2277:EML4_c.3172+744:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.668-3064:EML4_c.3172+29:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.668-4434:EML4_c.3172+407:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.668-1949:EML4_c.3173-171:ALKInv Protein Fusion: in frame {EML4:ALK}	
EVT-P-0005743-T02-IM5-245	P-0005743-T02-IM5		EML4-ALK	EML4	in-frame	three_prime	29447669	19	1058	True	retained	1.0	False	Protein kinase domain		EML4 NA (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 with ALK exons 20-29) : c.2243-181:EML4_c.3172+658:ALKInv Protein fusion: in frame (EML4-ALK) EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.2277:EML4_c.3172+744:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.668-3064:EML4_c.3172+29:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.668-4434:EML4_c.3172+407:ALKInv Protein Fusion: in frame {EML4:ALK}	
EVT-P-0045222-T03-IM7-246	P-0045222-T03-IM7		EML4-ALK	EML4	unknown	three_prime	29447583	19	1058	True	retained	1.0	False	Protein kinase domain		EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.2277:EML4_c.3172+744:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.668-3064:EML4_c.3172+29:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.668-4434:EML4_c.3172+407:ALKInv Protein Fusion: in frame {EML4:ALK}	
EVT-P-0045243-T01-IM6-247	P-0045243-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29448298	19	1058	True	retained	1.0	False	Protein kinase domain		EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.668-3064:EML4_c.3172+29:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.668-4434:EML4_c.3172+407:ALKInv Protein Fusion: in frame {EML4:ALK}	
EVT-P-0045341-T01-IM6-248	P-0045341-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29447920	19	1058	True	retained	1.0	False	Protein kinase domain		EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.668-4434:EML4_c.3172+407:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.668-1949:EML4_c.3173-171:ALKInv Protein Fusion: in frame {EML4:ALK}	
EVT-P-0065469-T02-IM7-249	P-0065469-T02-IM7		EML4-ALK	EML4	in-frame	three_prime	29446565	20	1058	True	retained	1.0	False	Protein kinase domain		EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.668-1949:EML4_c.3173-171:ALKInv Protein Fusion: in frame {EML4:ALK}	
EVT-P-0065210-T01-IM7-250	P-0065210-T01-IM7		EML4-ALK	EML4	in-frame	three_prime	29448076	19	1058	True	retained	1.0	False	Protein kinase domain		EML4 NA (NM_019063) - ALK (NM_004304) fusion: c.668-2012:EML4_c.3172+251:ALKInv Protein Fusion: in frame {EML4:ALK}	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	is_main_ref	is_disrupted	is_disrupted	description	annotation	effect_on_fra
EVT-P-0012873-T03-IM7-251	P-0012873-T03-IM7		EML4-ALK	EML4	in-frame	three_prime	29446439	20	1058	True	retained	1.0	False	Protein kinase domain		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+2231:45:ALKInv Protein	NA EML4_c.3173-	
EVT-P-0064716-T02-IM7-252	P-0064716-T02-IM7		EML4-ALK	EML4	in-frame	three_prime	29447739	19	1058	True	retained	1.0	False	Protein kinase domain		Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-3669:EML4 Protein	NA EML4_c.3172+588:ALKInv	
EVT-P-0031368-T04-IM6-253	P-0031368-T04-IM6		EML4-ALK	EML4	in-frame	three_prime	29447191	20	1058	True	retained	1.0	False	Protein kinase domain		Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-1154:EML4 Protein	NA EML4_c.3173-	
EVT-P-0045912-T02-IM6-254	P-0045912-T02-IM6		EML4-ALK	EML4	out-of-frame	three_prime	29446507	20	1058	True	retained	1.0	False	Protein kinase domain		Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.2154+1011:113:ALKInv Protein	NA EML4_c.3173-	
EVT-P-0031824-T01-IM6-255	P-0031824-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29450557	17	939	True	retained	1.0	False	Protein kinase domain		Fusion: out of frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+2971:19:ALKInv Protein	NA EML4_c.2816-	
EVT-P-0046083-T02-IM7-256	P-0046083-T02-IM7		EML4-ALK	EML4	in-frame	three_prime	29446665	20	1058	True	retained	1.0	False	Protein kinase domain		Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+986:271:ALKInv Protein	NA EML4_c.3173-	
EVT-P-0046594-T01-IM6-257	P-0046594-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29446445	20	1058	True	retained	1.0	False	Protein kinase domain		Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+2217:51:ALKInv Protein	NA EML4_c.3173-	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	is_main_ref	is_disrupted	is_fusion	is_inframe	is_disrupted	description	annotation	effect_on_fra
EVT-P-0040732-T02-IM6-258	P-0040732-T02-IM6		EML4-ALK	EML4	in-frame	three_prime	29446636	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-1832:EML4_c.3173-242:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+925:EML4_c.3173-215:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+925:EML4_c.3173-215:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+4042:EML4_c.3173-140:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-2813:EML4_c.3173-490:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+2231:EML4_c.3173-45:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2055:EML4_c.3173-697:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0047355-T01-IM6-259	P-0047355-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29446609	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+925:EML4_c.3173-215:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+925:EML4_c.3173-215:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+4042:EML4_c.3173-140:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-2813:EML4_c.3173-490:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+2231:EML4_c.3173-45:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2055:EML4_c.3173-697:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0047355-T02-IM6-260	P-0047355-T02-IM6		EML4-ALK	EML4	in-frame	three_prime	29446609	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+925:EML4_c.3173-215:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+4042:EML4_c.3173-140:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-2813:EML4_c.3173-490:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+2231:EML4_c.3173-45:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2055:EML4_c.3173-697:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0064342-T01-IM7-261	P-0064342-T01-IM7		EML4-ALK	EML4	in-frame	three_prime	29446534	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+4042:EML4_c.3173-140:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-2813:EML4_c.3173-490:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+2231:EML4_c.3173-45:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2055:EML4_c.3173-697:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0063585-T01-IM7-262	P-0063585-T01-IM7		EML4-ALK	EML4	in-frame	three_prime	29446884	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-2813:EML4_c.3173-490:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+2231:EML4_c.3173-45:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2055:EML4_c.3173-697:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0012873-T02-IM6-263	P-0012873-T02-IM6		EML4-ALK	EML4	in-frame	three_prime	29446439	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+2231:EML4_c.3173-45:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2055:EML4_c.3173-697:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0062999-T02-IM7-264	P-0062999-T02-IM7		EML4-ALK	EML4	in-frame	three_prime	29447091	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2055:EML4_c.3173-697:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	donor_frame	inframe	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0049318-T01-IM6-265	P-0049318-T01-IM6		EML4-ALK	EML4	unknown	three_prime	29446359	20	1070	False	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.2154+2335:EML4_c.3208:ALKInv Protein Fusion: mid-exon {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1641+1:EML4_c.3173-275:ALKInv Protein Fusion: out of frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+1166:EML4_c.3173-338:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.2242+9:EML4_c.3173-597:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2649:EML4_c.3173-964:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-639:EML4_c.3173-252:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4130:EML4_c.3173-380:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0046720-T01-IM6-266	P-0046720-T01-IM6		EML4-ALK	EML4	out-of-frame	three_prime	29446669	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.1641+1:EML4_c.3173-275:ALKInv Protein Fusion: out of frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+1166:EML4_c.3173-338:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.2242+9:EML4_c.3173-597:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2649:EML4_c.3173-964:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-639:EML4_c.3173-252:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4130:EML4_c.3173-380:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0049375-T02-IM6-267	P-0049375-T02-IM6		EML4-ALK	EML4	in-frame	three_prime	29446732	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+1166:EML4_c.3173-338:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.2242+9:EML4_c.3173-597:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2649:EML4_c.3173-964:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-639:EML4_c.3173-252:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4130:EML4_c.3173-380:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0049511-T01-IM6-268	P-0049511-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29446991	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.2242+9:EML4_c.3173-597:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2649:EML4_c.3173-964:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-639:EML4_c.3173-252:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4130:EML4_c.3173-380:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0010177-T02-IM7-269	P-0010177-T02-IM7		EML4-ALK	EML4	in-frame	three_prime	29447358	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2649:EML4_c.3173-964:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-639:EML4_c.3173-252:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4130:EML4_c.3173-380:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0031827-T03-IM6-270	P-0031827-T03-IM6		EML4-ALK	EML4	in-frame	three_prime	29446646	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-639:EML4_c.3173-252:ALKInv Protein Fusion: in frame {EML4:ALK} EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4130:EML4_c.3173-380:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	
EVT-P-0051231-T01-IM6-271	P-0051231-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29446774	20	1058	True	retained	1.0	False	Protein kinase domain				EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4130:EML4_c.3173-380:ALKInv Protein Fusion: in frame {EML4:ALK}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint	breakpoint	breakpoint	protein	protein	breakpoint	main_ref	chr	start	end	domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0060148-T01-IM7-279	P-0060148-T01-IM7		EML4-ALK	EML4	in-frame	three_prime	29447757	19	1058	True	retained	1.0	False	Protein kinase domain					EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+2600: EML4_c.3172+570: ALKInv	NA	
EVT-P-0059699-T03-IM7-280	P-0059699-T03-IM7		EML4-ALK	EML4	in-frame	three_prime	29447467	19	1058	True	retained	1.0	False	Protein kinase domain					EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-1990: EML4_c.3172+860: ALKInv	NA	
EVT-P-0031827-T06-IM7-281	P-0031827-T06-IM7		EML4-ALK	EML4	in-frame	three_prime	29446646	20	1058	True	retained	1.0	False	Protein kinase domain					EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-639: EML4_c.3173-252: ALKInv	NA	
EVT-P-0054263-T01-IM6-282	P-0054263-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29446557	20	1058	True	retained	1.0	False	Protein kinase domain					EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-210: EML4_c.3173-163: ALKInv	NA	
EVT-P-0059257-T02-IM7-283	P-0059257-T02-IM7		EML4-ALK	EML4	in-frame	three_prime	29448210	19	1058	True	retained	1.0	False	Protein kinase domain					EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-982: EML4_c.3172+117: ALKInv	NA	
EVT-P-0054408-T01-IM6-284	P-0054408-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29446902	20	1058	True	retained	1.0	False	Protein kinase domain					EML4 (NM_019063) - ALK (NM_004304) fusion: c.2243-262: EML4_c.3173-508: ALKInv	NA	
EVT-P-0008332-T02-IM6-285	P-0008332-T02-IM6		EML4-ALK	EML4	unknown	three_prime	29446913	20	1058	True	retained	1.0	False	Protein kinase domain					EML4 (NM_001145076) - ALK (NM_004304) fusion: c.1315+828: EML4_c.3173-519: ALKdup	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	breakpoint_4	main_ref	chr1_start	chr1_end	chr2_start	chr2_end	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0005711-T02-IM6-292	P-0005711-T02-IM6		KIF5B-ALK	KIF5B	in-frame	three_prime	29449548	18	1023	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - ALK (NM_004304) fusion (KIF5B exons 1-24 fused with ALK exons 19-29) : t(2;10)(p23.2;p11.22)(chr2:g.29449548:::)	NA	
EVT-P-0008764-T01-IM5-293	P-0008764-T01-IM5		KIF5B-ALK	KIF5B	unknown	three_prime	29448327	19	1058	False	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - ALK (NM_004304) fusion (KIF5B exons 1-24 fused with ALK exons 19-29) : t(2;10)(p23.2;p11.22)(chr2:g.29448327:::)	NA	
EVT-P-0017924-T01-IM6-294	P-0017924-T01-IM6		KIF5B-ALK	KIF5B	in-frame	three_prime	29447945	19	1058	True	retained	1.0	False	Protein kinase domain				KIF5B (NM_004521) - ALK (NM_004304) fusion : t(2;10)(p23.2;p11.22)(chr2:g.29447945:::)	NA	
EVT-P-0008764-T02-IM6-295	P-0008764-T02-IM6		KIF5B-ALK	KIF5B	unknown	three_prime	29448327	19	1058	False	retained	1.0	False	Protein kinase domain				Fusion {KIF5B:ALK} (KIF5B (NM_004521) - ALK (NM_004304) Fusion (KIF5B exons 1-24 fused with ALK exons 19-29) : t(2;10)(p23.2;p11.22)(chr2:g.29448327:::)	NA	
EVT-P-0067512-T01-IM7-297	P-0067512-T01-IM7		NOTCH1-ALK	NOTCH1	in-frame	three_prime	29449152	18	1023	True	retained	1.0	False	Protein kinase domain				NOTCH1 (NM_017617) - ALK (NM_004304) fusion : t(2;9)(p23.2;q34.3)(chr2:g.29449152:::)	NA	
EVT-P-0034044-T01-IM6-298	P-0034044-T01-IM6		NRP2-ALK	NRP2	in-frame	three_prime	29448550	19	1023	True	retained	1.0	False	Protein kinase domain				in frame {NOTCH1:ALK} NRP2 (NM_201266) - ALK (NM_004304) rearrangement: c.1291+575_c.3068-119inv Protein Fusion: in frame {NRP2:ALK}	NA	

3 Figures

ALK domain-retention track

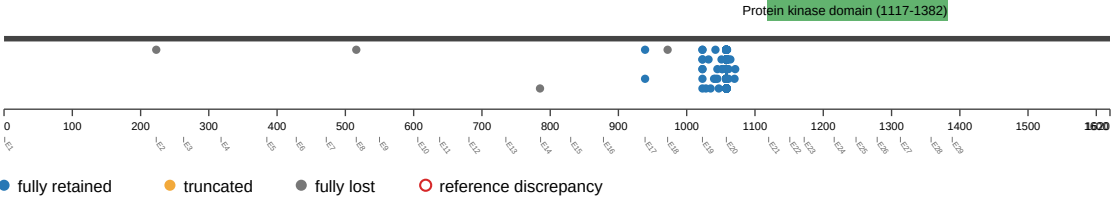
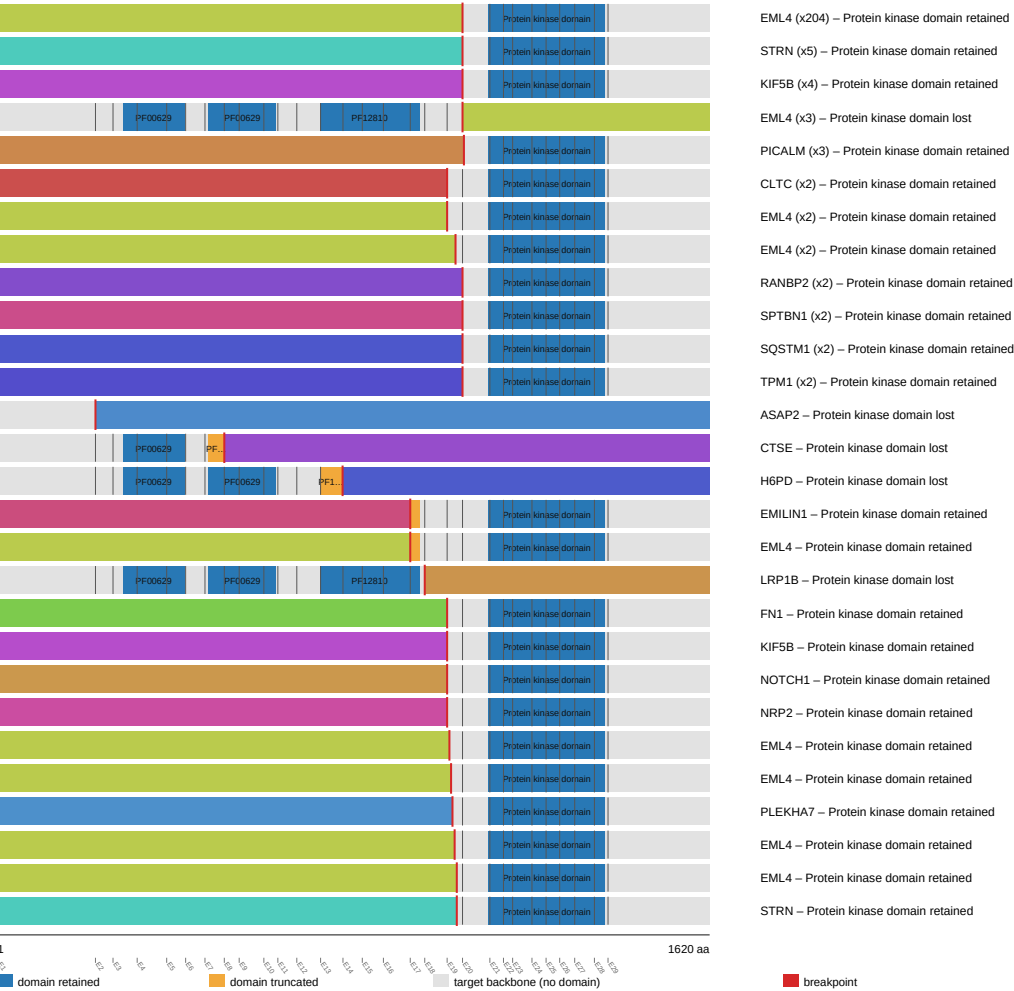


Figure 1: domain retention outliers

ALK fusion-transcript schematic

partner block → breakpoint → retained ALK portion (domain-colored, exon ticks)



Showing the top 28 of 50 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

Figure 2: fusion schematic

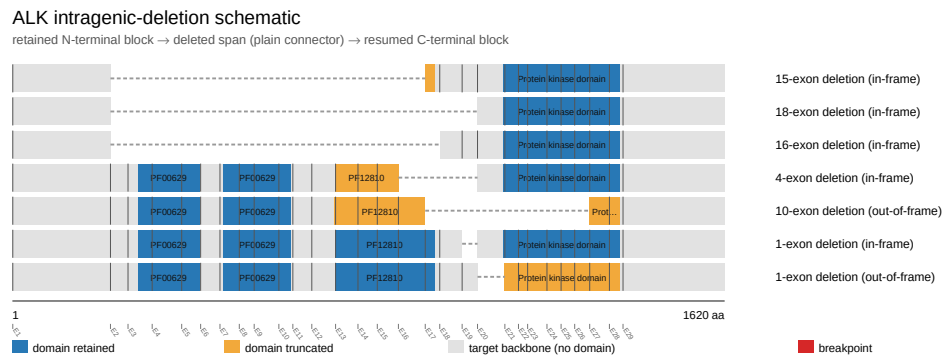


Figure 3: intragenic deletion schematic

Reference vs msk_impact_50k_2026



Figure 4: reference comparison